

SYNOPSIS

At 0738 local time on 21 June 2004, the 74,373gt UK registered container vessel, *Hyundai Dominion*, and the 6,899gt Hong Kong registered container vessel, *Sky Hope*, collided in the East China Sea. There were no injuries or pollution. Only *Sky Hope* suffered any significant damage. Each vessel was able to continue passage.

As the vessels approached, in good visibility, the officer on watch on *Sky Hope* incorrectly assessed the encounter as one where *Hyundai Dominion* was overtaking his vessel. Action by either vessel was then delayed by discussions on the VHF. Further delay resulted when the OOW on *Hyundai Dominion* requested the other vessel to keep clear using the free text facility on the Automatic Identification System (AIS).

In spite of very late avoiding action taken by both vessels, they collided. The starboard bridge wing, lifeboat davit and a container on *Sky Hope* were damaged. Damage to *Hyundai Dominion* was limited to slight indentation of a breakwater on her port bow, distorted handrails on the forward deck and scratches to paintwork on the port bow.

The accident was investigated by the UK's Marine Accident Investigation Branch (MAIB) and the Hong Kong Marine Department (HKMD) as a joint investigation in accordance with the International Maritime Organization's (IMO's) Code for the investigation of marine casualties and incidents.

The investigation highlighted several causal and contributory factors. These included:

- Neither watchkeeper claimed to be fatigued, however both had worked in excess of the hours permitted under the international convention on Standards of Training, Certification and Watchkeeping for seafarers (STCW) over the previous two days.
- *Sky Hope* had been observing the approach of *Hyundai Dominion*. However, other than VHF communication there was no avoidance action taken until she was within a range of 0.2 nautical mile (nm).
- *Sky Hope* judged *Hyundai Dominion* to be an overtaking vessel which, in accordance with the COLREGS, required him to take no immediate avoiding action.
- *Hyundai Dominion* considered *Sky Hope* was a crossing vessel requiring *Hyundai Dominion* to "stand-on".
- In VHF communications between the vessels leading up to the collision, it is likely that a disagreement took place due to the difference in opinion over the "crossing" or "overtaking" situation.
- The OOW of *Hyundai Dominion* stated that he sent a text message over AIS asking *Sky Hope* to keep clear. The OOW of *Sky Hope* stated that he did not receive this message.
- *Hyundai Dominion* made a sound signal before the collision, using the forward whistle. There was no sound signal given by *Sky Hope* before the collision.

- Neither OOW advised his master prior to the collision.
- Correct emergency procedures were not followed by *Hyundai Dominion* after the collision.
- *Sky Hope* resumed passage some 22 minutes after the collision. It is unlikely that a proper assessment of the vessel's condition could have been completed within this time.
- The bridge watchkeepers of *Hyundai Dominion* lacked a clear understanding of the operation of the engine controls.

Following the collision, the managers of *Hyundai Dominion* have issued the results of a review of company navigational procedures, which began in May 2004. They have also introduced company specific navigational training for junior officers and deck cadets. In addition, they have increased the number of internal and external navigational audits of their vessels.

Recommendations have been made to the managers of both vessels to advise their bridge watchkeepers to call the vessel's master at the early stages of a developing hazardous situation, the importance of ensuring watchkeepers receive adequate rest and the procedures to be followed in the event of a collision. Further recommendations have been made to the managers of *Sky Hope* regarding application of the COLREGS and use of VHF and sound signals in collision avoidance. Recommendations have been made to *Hyundai Dominion*'s managers with respect to familiarisation of bridge watchkeepers with engine controls, the use of AIS text facilities in situations requiring prompt action and the need for OOWs to be able to communicate with other bridge team members.

Similarly, recommendations have been made to the International Chamber of Shipping (ICS) to promulgate to its members the lessons learned from this accident regarding the dangers of using AIS text facilities in situations requiring prompt action. It is also recommended to reinforce the advice contained in the ICS's *Bridge Procedures Guide* covering use of the COLREGS, calling the vessel's master and post collision actions.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *HYUNDAI DOMINION* AND *SKY HOPE* AND ACCIDENT

Hyundai Dominion (Figure 1)

Vessel details

Registered owner	:	Tempo Navigation Ltd
Manager(s)	:	Zodiac Maritime Agencies Ltd Lynton House 7-12 Tavistock Square London WC1H 9TP
Port of registry	:	London
Flag	:	UK
Type	:	Container
Built	:	2001, Hyundai Heavy Industries, Ulsan, Korea
Classification society	:	Lloyd's Register
Construction	:	Steel
Length overall	:	303.83m
Gross tonnage	:	74,373
Engine power and type	:	65930kW. B&W 12K98MC-CX
Service speed	:	26.4knots
Other relevant info	:	Single fixed pitch propeller
Persons on board	:	22
Damage	:	Minor damage to guardrails and breakwater on the port bow.

***Sky Hope* (Figure 2)**

Vessel details

Registered owner	:	Topwave Shipping Ltd
Manager(s)	:	Pagasa Shipmanagement Inc
Port of registry	:	Hong Kong
Flag	:	Hong Kong
Type	:	Container
Built	:	2000, Shin Kurushima Dockyard Co Ltd, Akitsu, Japan
Classification society	:	Nippon Kaiji Kyokai
Construction	:	Steel
Length overall	:	120.84m
Gross tonnage	:	6,899
Engine power and type	:	5,589kW, B&W 8S35MC
Service speed	:	15.6knots
Other relevant info	:	Single fixed pitch propeller
Persons on board	:	18
Damage	:	Significant damage to starboard bridge wing, deck edges and guardrails, freefall lifeboat and davit and one cargo container

Accident details

Time and date	:	0738LT, 21 June
Location of accident	:	30° 59.7'N 125° 45.7'E (Figure 3)
Injuries/fatalities	:	None

1.2 NARRATIVE

All courses are true, all times are local (UTC+9)

1.2.1 Events as seen by *Hyundai Dominion*

Hyundai Dominion sailed from Kaosiung, Taiwan, on 20 June for passage to Pusan, Korea. Between 0710 and 0720 the following day, the officer of the watch (OOW), who was the chief officer, detected a vessel on his port bow at a range in excess of 5nm. The vessel was initially detected by radar but sighted immediately after. *Hyundai Dominion* was on a course of 036° in autopilot, at a speed of 22 knots. An able seaman (AB) was on watch on the bridge with the chief officer, but he temporarily left the bridge at about 0700.

The radar contact was identified as *Sky Hope* using the information displayed by the AIS. The automatic radar plotting aid (ARPA) on the port radar display was the display chiefly used by the OOW and was set to the 12-mile range scale, off-centred to the south-west. The starboard radar display was on the 6-mile range scale, and was also off-centred to the south-west. Both radar displays were north up and operating in true motion. The OOW assessed it to be a crossing situation and expected the other vessel to alter course to starboard in order to keep clear.

Between 0725 and 0730, the range of *Sky Hope* had closed to about 2.5 miles, and the closest point of approach (CPA) of *Sky Hope* was 0.3 mile by ARPA. The OOW switched the port radar display to relative motion, and used the relative vector of *Sky Hope* to determine that she would pass ahead. He was aware that the ARPA information did not take into account the distance between the radar aerial and the bow, and considered 0.3 mile to be too close. The OOW called *Sky Hope* by very high frequency (VHF) radio channel 16, but did not hear a response.

Two to three minutes later, the OOW sent a message to *Sky Hope* via AIS, stating:

PLS KEEP CLEAR

AIS indicated that the message had been successfully transmitted, but no reply was received from *Sky Hope*. About 1.5 to 2 minutes later, the OOW decided to call again via VHF radio channel 16, and on this occasion there was an immediate response. The OOW of *Sky Hope* acknowledged that he had received the AIS message and stated an intention to pass ahead of *Hyundai Dominion* because this was an overtaking situation. The OOW of *Hyundai Dominion* told him not to cross his bows, but to alter course to starboard, as this was a crossing, not an overtaking situation. The OOW of *Sky Hope* agreed to take this action.

By that stage the AB had returned to the bridge and, as a precaution, the OOW of *Hyundai Dominion* switched to manual steering. Moments later, the OOW received a call from *Sky Hope* to say that it was too late to alter course to starboard, and he was altering to port. *Hyundai Dominion*'s OOW replied that he would alter to starboard to help the situation. He ordered the AB to put the helm hard to starboard and checked the rudder angle indicator to ensure this instruction had been carried out correctly. One short blast was also sounded using the forward whistle only. By then, *Sky Hope* was disappearing from view behind the containers on the port bow, and the OOW continued to watch her approach on a closed circuit television (CCTV) monitor on the port side of the bridge, which had a camera on the port bridge wing selected. The rate of turn indicator was showing 30° per minute, and *Hyundai Dominion* was heeling to port.

The two vessels collided at about 0738, with *Hyundai Dominion*'s port bow hitting *Sky Hope*'s starboard quarter. About 10 seconds after the collision, the OOW alerted the master in his cabin using an alarm fitted to the internal telephone system. The master had not felt the ship heel during the turn to starboard, heard the one short blast on the whistle, or felt the impact of the collision.

When the master arrived on the bridge, *Sky Hope* was already between 200 and 300 metres on the port quarter, but he could see that her starboard bridge wing had been damaged. The chief officer had already steadied on a course of 069° but had not altered speed. The master called both the second and third officers in their cabins by telephone, and ordered them to the bridge. On their arrival, the third officer took over the watch and the second officer assisted the master. The master did not sound the general alarm. Information was exchanged with *Sky Hope*. The chief officer checked the contents of the ballast tanks using the control/monitoring system on the bridge, and also ordered the bosun to take soundings. At about 0757, the master ordered the speed to be reduced, and at about 0820, course was altered to about 013° to regain the planned track. He did not consider that there was any need to reverse course.

1.2.2 Events as seen by *Sky Hope*

At about 0715 on 21 June 2004, while underway from Shanghai to Osaka and Kobe, the OOW of *Sky Hope* sighted a large container ship at her starboard quarter, which was later identified as *Hyundai Dominion*. The course and speed of *Sky Hope* was 091° and 15.3 knots.

At 0720, the bearing and range of the container ship was found to be 210° at 6nm. The chief officer, who was the OOW, did not take any avoiding action at that stage as he assessed that this was an overtaking situation and *Hyundai Dominion* should keep clear.

As *Hyundai Dominion* appeared to be taking no action, the OOW shifted to manual steering at 0725. At about 0730, the OOW observed on radar that the bearing of *Hyundai Dominion* had not changed and the distance had decreased to 3nm. The OOW called *Hyundai Dominion* on VHF radio Channel 16 at 0730, 0732, and 0734 and told her to keep clear. However, *Hyundai Dominion* did not appear to have taken any action and continued to approach from the starboard quarter, with no change of bearing.

At 0736, at a distance of about 0.2nm, the OOW altered course to 065° to try and avoid a collision. The collision occurred at about 0738, as the port bow of *Hyundai Dominion* struck the starboard quarter of *Sky Hope* in the region of the lifeboat installation, advancing its impact towards the starboard wing of the navigational bridge and into the container stowage bay area. *Hyundai Dominion* did not appear to have slowed down, and it disengaged from *Sky Hope* after the impact. *Sky Hope*'s OOW called up on the VHF Channel 16 after the collision and exchanged information.

After the collision, *Sky Hope*'s master raised the general alarm. The crew was mustered to emergency stations. After checking there were no casualties, it was found that *Sky Hope*'s starboard quarter had sustained extensive damage during the impact. *Hyundai Dominion* confirmed to *Sky Hope* that there was no serious damage to the vessel or injury to her crew. The master assessed the damage, and confirmed that the ship's hull remained intact.

Sky Hope resumed her voyage to Osaka and Kobe at 0800.

1.3 WEATHER CONDITIONS

Visibility was good, wind was north-west force 5, and the sea state was moderate. Sunrise was at 0533 (local time).

1.4 DAMAGE

1.4.1 *Hyundai Dominion*

The damage to *Hyundai Dominion* was limited to slight indentation of a breakwater on her port bow, distorted handrails on the forward deck, and scratches to the paintwork on her port bow (**Figures 4 and 5**).

1.4.2 *Sky Hope*

The damage sustained by *Sky Hope* was more extensive and included: the starboard bridge wing crushed inwards; starboard "C" and "D" deck railings crushed or bent and the deck corners badly dented; the davit arm for the freefall lifeboat badly crushed and the lifeboat shell cracked; one liferaft damaged; one cargo container crushed (**Figures 6, 7, 8 and 9**).

SECTION 2 - ANALYSIS

Times shown are local time (UTC+9)

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

Section A VIII/1 of the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended in 1995 (STCW) with respect to fitness for duty, states:

- “1. All persons who are assigned duty as officer in charge of a watch or as a rating forming part of a watch shall be provided a minimum of 10 hours of rest in any 24-hour period.*
- 2. The hours of rest may be divided into no more than two periods, one of which shall be at least 6 hours in length.*
- 3. The requirements for rest periods laid down in paragraphs 1 & 2 need not be maintained in the case of an emergency or drill or in other overriding operational conditions.*
- 4. Notwithstanding the provisions of paragraphs 1 & 2, the minimum period of ten hours may be reduced to not less than 6 consecutive hours provided that any such reduction shall not extend beyond two days and not less than 70 hours of rest are provided each seven-day period.”*

The watchkeepers on both vessels had 8 hours available for rest before they began their watches on the morning of the collision. However, both had worked in excess of the hours permitted by STCW over the previous 2 days.

Neither watchkeeper claims to have been affected by fatigue; this is not uncommon following an accident. The managers of both vessels should, however, take note of the breach of STCW hours of work. Due to this breach, it is impossible to dismiss fatigue as being a contributory factor to the collision. Both watchkeepers were probably affected to a similar degree. Again, the managers of both vessels should recognise that fatigue is likely to adversely affect the performance of their staff when they are unable to take the hours of rest specified in STCW and should take remedial action.

2.3 RECORDED EVENTS BEFORE COLLISION

Summary of observations from each vessel

Sky Hope was on a course of 091° at a speed of 15.3 knots. Her OOW first saw *Hyundai Dominion* at 0720 at 6nm on a bearing of 210°. He assessed that *Hyundai Dominion* was an overtaking vessel, and that *Sky Hope* was the stand on vessel, and as such, was not required to take any immediate avoiding action. At 0730, the range had closed to 3nm but the bearing remained unchanged. At 0736, the OOW of *Sky Hope* estimated that *Hyundai Dominion* had closed to 0.2nm, and he altered course to port to 065° in an attempt to avoid a collision.

Hyundai Dominion was on a course of 036° at a speed of 22 knots. Her OOW first saw *Sky Hope* between 0710 and 0720 at 45° on his port bow, and at a range in excess of 5nm. He assessed that the ships were in a crossing situation in which *Sky Hope* was the give way vessel, and that *Hyundai Dominion* was the stand on vessel. Between 0725 and 0730, the OOW saw that the range had decreased to 2.5nm but her bearing remained unaltered. He also checked the CPA by ARPA to be 0.3nm ahead. At 0736, the OOW of *Hyundai Dominion* applied the helm hard to starboard in order to avoid colliding with *Sky Hope*. Although the ship started to turn to starboard, the two vessels collided between 0738 and 0740.

Other than VHF communication, neither OOW took any action to avoid a collision until the ships were in very close proximity, because both were under the impression that they were the stand on vessel. As the situation could not involve the ships overtaking and crossing at the same time, there is clearly a conflict between the two accounts.

2.4 CLASSIFICATION OF OVERTAKING OR CROSSING OF TWO VESSELS

In an effort to resolve the conflicting accounts of the OOWs, the recorded speed and course for each vessel has been plotted to find the resultant relative vector.

Figure 10 shows that *Sky Hope* closed *Hyundai Dominion* from a bearing of 353°, and that *Hyundai Dominion* closed *Sky Hope* from a bearing of 173°. The closing speed of the two vessels was about 18 knots.

This relative vector has been transferred through the initial ranges and bearings of the other vessel, as observed by each OOW (**Figures 11 and 12**).

Figure 11 shows that if *Hyundai Dominion* was first seen by *Sky Hope*'s OOW on a bearing of 210° at 6 miles, she would have passed 3.6nm astern of *Sky Hope*. There would therefore have been no risk of collision.

Figure 12 shows that if *Sky Hope* was first seen by the OOW on board *Hyundai Dominion* 45° on the port bow (351° true), at about 5nm, *Hyundai Dominion* would have passed in the order of 0.5 cable ahead of *Sky Hope*. Given the

smaller CPA produced by this plot, along with the fact that the reported visual bearing (351°) was within 2° of the predicted bearing (353°), it is considered that the account of the OOW on board *Hyundai Dominion* is more accurate with regard to the relative positions of the vessels.

Figure 13 is a representation of Rules 13 and 15 of the International Regulations for Preventing Collisions at Sea 1972 as amended (COLREGS).

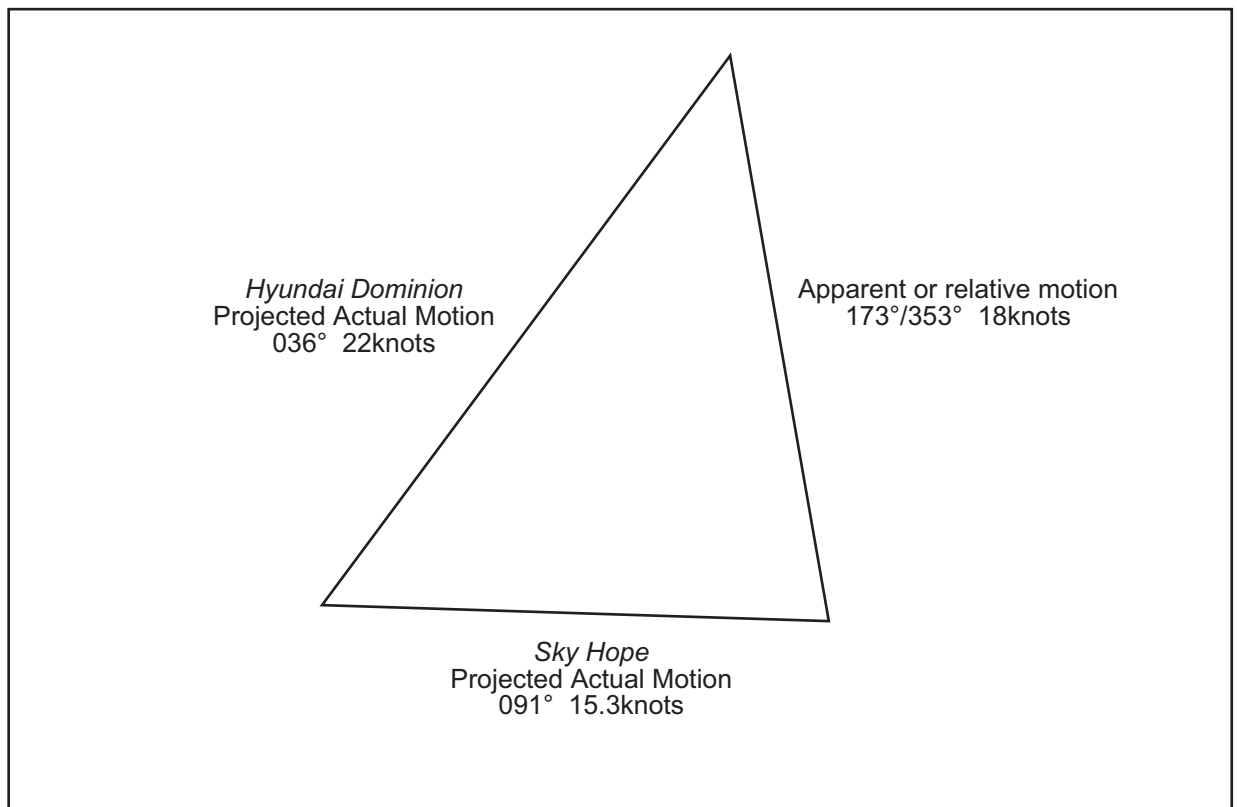
A vessel shall be deemed to be overtaking when approaching another vessel from a direction more than 22.5° abaft her beam. Otherwise, the two vessels shall be crossing.

For *Hyundai Dominion* to have been an overtaking vessel, she needed to approach *Sky Hope* from an angle greater than 22.5° aft of the beam. As *Sky Hope* was on a heading of 091°, *Hyundai Dominion* needed to be approaching at an angle greater than $91^\circ + 112.5^\circ$ (or greater than 203.5°). In other words, if *Hyundai Dominion* was observed at 0720 on a bearing of 210°, it was 7° or 8° behind prescribed limit and, thus, could be regarded as an overtaking vessel.

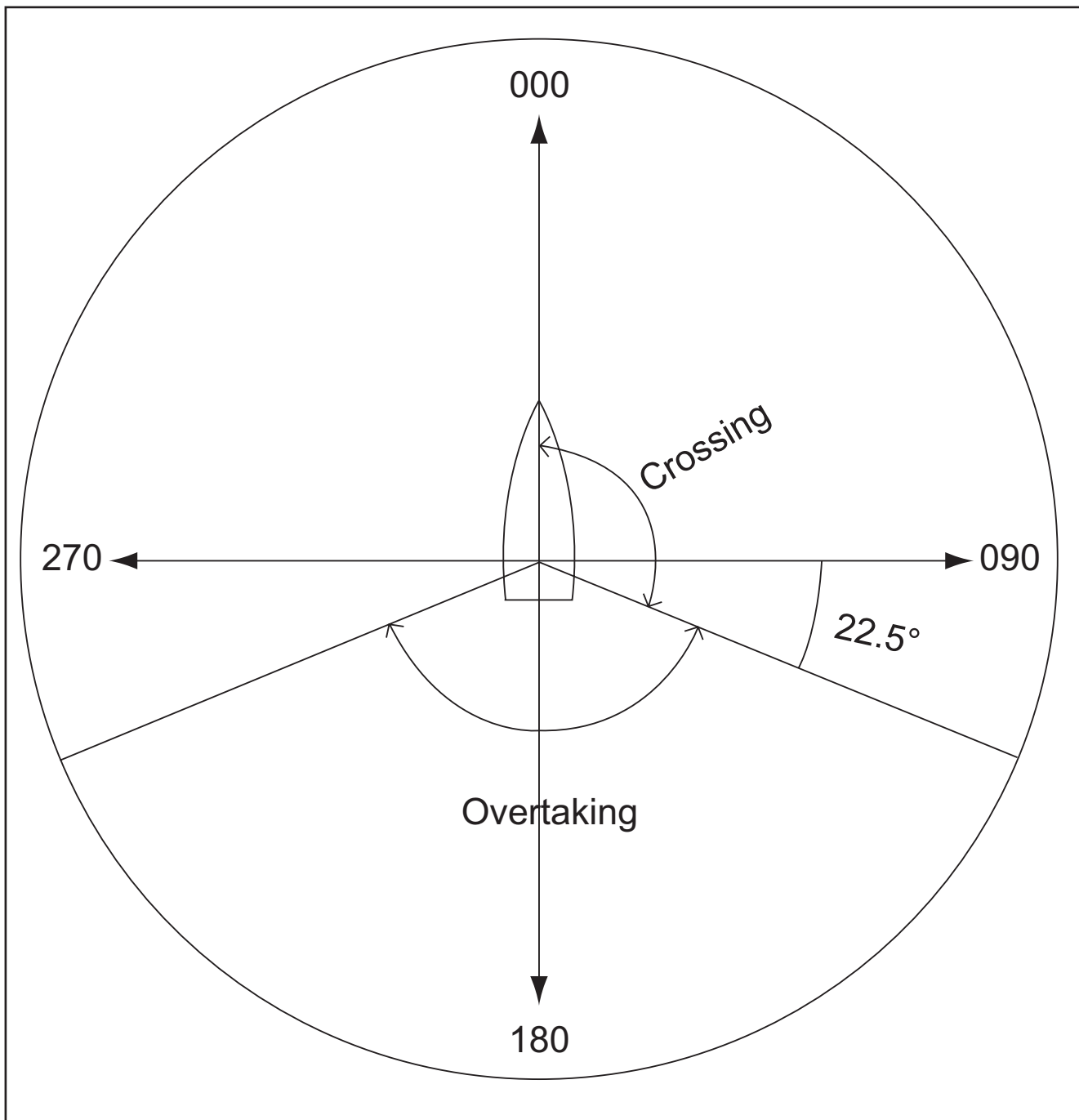
However, as **Figure 10** shows, *Hyundai Dominion* closed on a bearing of 173°, not 210°.

As 173° was forward of the starboard beam of *Sky Hope*, this was a crossing, not an overtaking situation as assessed by *Sky Hope*'s OOW.

Figure 10



Plot to find relative motion



Representation of Rules 13 and 15 (COLREGS)

2.5 THE COLLISION

It has not been possible to establish how the OOW of *Sky Hope* obtained the range and bearing of 6 miles and 210°, but it is apparent this was an inaccurate observation that resulted in him assessing *Hyundai Dominion* as an overtaking vessel. This contributed to the collision. As the crossing vessel, *Sky Hope* was the give way vessel in accordance with Rule 15 of the COLREGS, and under Rule 16 she should have taken early and substantial action to keep out of the way of *Hyundai Dominion* (**see Annex**).

The OOW of *Hyundai Dominion* considered his own ship was the stand-on vessel, which appears to have been correct. However, this did not prevent him from taking earlier avoiding action in accordance with Rule 17(a)(ii) of the COLREGS (**see Annex**) when he found that *Sky Hope* was not taking the appropriate action. *Hyundai Dominion* did not take action to alter course until 0736, when *Sky Hope* was in such close range that any action taken by the latter vessel would not have prevented the collision. This is considered to be in accordance with Rule 17(b) of the COLREGS.

The avoiding actions taken by the two vessels are considered to have been taken far too late given the size and close proximity of these vessels. Given the open sea room, and the manoeuvring characteristics of the two vessels, there is no reasonable explanation as to why the two OOWs did not take avoiding action earlier. Had substantial avoiding action been taken earlier, by either vessel, the collision would have been avoided.

The decision by both OOWs not to take action until too late might have been influenced by their fatigue.

2.6 SPEED

It is apparent that at no stage did the OOW on *Hyundai Dominion* consider reducing speed in an effort to avoid the collision, although it is accepted that this might not have been the most appropriate action. However, it is of concern that he was so uncertain of the proper use of the engine controls, and the consequences of their movement, that a speed reduction was not on his list of options, either for collision avoidance or as a post collision action.

The other bridge watchkeeping officers on *Hyundai Dominion* shared a similar lack of knowledge of the use of the main engine controls at sea. This problem had previously been identified by the vessel's managers who had issued an explanatory fleet circular on the subject. Clearly, with respect to the bridge watchkeeping officers serving on board *Hyundai Dominion*, this initiative had failed and the issue needs further review by the managers.

The speed of *Sky Hope* was not altered before the collision. Although she altered course at about 0736 in an effort to avert collision, she maintained her speed of 15.3 knots.

The OOW on *Sky Hope* considered that *Hyundai Dominion* was an overtaking vessel, and judged that a speed reduction would worsen the closing situation between the two vessels.

SECTION 3 - CONCLUSIONS

The following safety issues have been identified by the investigation. They are not listed in any order of priority.

3.1 FINDINGS

1. The failure of both watchkeepers to take early action to avoid a collision was the cause of this accident. However, it must be recognised that both men had worked hours in excess of those prescribed under STCW. Consequently, fatigue is considered to have made some contribution to their poor decision making and judgment, albeit to an uncertain degree. [2.2]
2. *Sky Hope* was, according to the COLREGS, clearly the give way vessel in this encounter. No single clear cause has been found to explain the OOW's misjudgment of his vessel's status. [2.4]
3. Although she was the stand-on vessel, *Hyundai Dominion* still had the opportunity to take action in good time when it was seen that *Sky Hope* had taken no action. Rather than take positive and early action, the OOW wasted time in sending an AIS text message and in VHF discussions with the other vessel. These were unnecessary and time consuming diversions. Guidance has been issued to the industry for the use of AIS at sea, but this accident suggests this needs to be reinforced, particularly as these are systems that are very new to seafarers. [2.8]
4. The OOW on neither vessel called the master before the collision. In view of the almost universal standing instructions given by masters, the requirements of STCW and guidance given during the formal training of navigating officers, this must be a matter of concern. The managers of both vessels should reinforce their requirements on this matter. [2.10]

3.2 OTHER FINDINGS

1. Reducing speed to avoid this collision was probably not appropriate in this case for the OOW on *Hyundai Dominion*. Speed reduction should, however, always be an option that is open to any bridge watchkeeper. Clearly the watchkeeper needs to be competent in operating the engine controls and understand the likely results of doing so. This was not the case for the OOW on *Hyundai Dominion*, or for his two colleagues, all of whom had an uncertain understanding of the controls. In view of this, none considered speed reduction as an option for dealing with a situation at sea. The vessel's managers needed to address this subject following an earlier accident with one of its vessels. The issue needs to be reviewed once more. [2.6]
2. A number of recent investigations undertaken by the MAIB have shown that vessels' general alarms are frequently not sounded during or after an accident. This was so on *Hyundai Dominion*, and might be related to lack of clarity in the

emergency procedures. The absence of an emergency signal could mean that crew are not alerted to an actual or potential emergency, and are not given the necessary sense of urgency to act. The managers of *Hyundai Dominion* should clarify their guidance to masters on their expectations concerning the sounding of the general alarm. [2.11]

3. Although the general alarm was sounded on *Sky Hope*, the vessel resumed her passage only 22 minutes after the collision. It is difficult to believe that all the necessary safety checks could have been performed in this time. Resuming the passage might have aggravated undetected damage. The owners of *Sky Hope* should clarify their instructions to their masters on the precautions to be taken following a collision. [2.12]
4. The managers of *Hyundai Dominion* could usefully enhance their safety management system by making historical navigational data available during navigational audits. The audits presently undertaken appear to be of a good standard, but their value could be significantly enhanced if such data was available to auditors. To do this, the managers would need to arrange for their vessels' electronic chart systems to be set to the record mode. [2.13]
5. Good communications are important in the safe operation of any ship, no more so than between watchkeepers on the same vessel. The chief officer and AB lookout on *Hyundai Dominion* were unable to communicate using a common language. This is a matter of concern, which, under many foreseeable circumstances, could result in the aggravation of difficult and dangerous situations. The policy of using watchkeepers on any vessel who are unable to communicate with each other is a matter for concern that should be addressed. [2.14]